BANL6900 Business Analytics Capstone

Final Project Paper Report

Xoxoday.com: Customer Engagement through Social Media

Nandini Kovuru

Akhila Thakur

Madhurima Dilari

Shiva Shankar Gundla

Preethi Vudugula

University of New Haven

12/05/2025

1. Abstract:

In this project, the researcher explores the dynamics of the user interaction in the Twitter network of the Xoxoday in order to comprehend how the process of social interaction, power, and dissemination of information takes place within the brand environment. The project bases its analysis on the use of social network analysis, analyzing both mentions, replies, and retweets to identify structural properties of engagement and what drives or constrains organic audience engagement. The data in the present study will consist of mentions-and-replies interactions, retweet relationships, and metadata mined out of Xoxoday-related tweets and complemented by interactive Power BI visualizations and graph metrics created in R.

The research uses descriptive social network analysis techniques to calculate centrality, find influential accounts, and discover communities and gauge the level of engagement with time. Based on the findings, it is shown that the Twitter network is highly centralized on the brand account, that the level of engagement significantly depends on contest-based user actions, and that viral events rely on isolated cases. The project ends with some strategic conclusions of how to enhance the engagement of Xoxoday in the long-term and suggestions of how to strengthen the network by applying more diversified and genuine interaction patterns.

2. Introduction:

Companies are relying on social media like Twitter to create brand awareness, customer relationship and engagement. In the case of Xoxoday, a company in the rewards and employee engagement industry, Twitter is an effective medium of engaging customers, content promotion, and amplification of brand story. Yet, even though the posting was active, there had been no systematic evaluation of the actual engagement patterns, influence distribution, as well as depth of conversations by Xoxadays. It is hard to work out an efficient social media strategy, enhancing the presence of the community and diffusion of the message without knowing how people engage with the brand and among themselves in the first place. The business case discussed in the project is the necessity to assess the quality of the social media engagement of Xoxoday and whether the existing interactions on Twitter can benefit the sustainable expansion of its audience. Through examining the network of mentions, replies, and retweets, it is hoped to determine whether the engagement ecosystem is healthy, diverse, and organic at Xoxoday or whether it is excessively reliant upon particular campaigns or artificial tagging, or other isolated viral moments. The resolution of this issue will give Xoxoday some facts about how to better target the audience, minimize the number of interactions with poor quality, and create a more stable online community.

According to previous studies, the analysis of social networks is an essential means of realizing digital engagement tendencies and information circulation in online communities. Other studies like Borgatti et al. (2009) underline that the network centrality measures are beneficial in identifying the influential actors as well as the communication bottlenecks in the organizational and social settings. Studies conducted by Kwak et al. (2010) also indicate that Twitter networks can be highly centralized with only a few accounts controlling the attention, thus the diffusion of information as well as restricting the interaction between peers. This coincides with the worries that brand oriented communication is not likely to make a sustainable engagement given that it is not backed by distributed influence. The pattern of engagement has been addressed by other scholars on how user-generated content and viral marketing influence the pattern of engagement. According to Berger and Milkman (2012), emotional and high-arousal messages are more likely to be shared and this explains the reason why some tweets become viral and others have little attention. Secondly, Liu and Dai (2020) show that contest-based or incentivized tagging behavior can cause artificial spikes in activity that is not an actual interest of the audience, which is especially concerned with the behavior of Xoxadays data. All these studies lead to the conclusion that engagement should be assessed by considering the differences between natural interaction, fake boosting, and the strategic value of influential nodes to the process of creating online communities.

This report is the extensive Social Network Analysis (SNA) of the Twitter communications of Xoxoday in terms of mentions/replies and retweets. Computational graph techniques were utilized to analyze the structural relationships between the users, centrality distributions, community formation, and temporal trends of interaction and backed with uploaded visualizations. These results indicate that there are various trends concerning the concentration of engagement, artificial and frivolous inflation due to the activity of the contest, and reliance on isolated viral experiences.

3. Data Sources and Analytical Approach:

The node and edge files were analyzed on both the mentions/replies and retweets networks which were the Tweets and the Retweets. Igraph, tidygraph and ggraph created graphs and metrics in R, and the findings in Power BI were used to supplement the structural results. Images which have been uploaded include network visualizations and statistical summaries, which have been interpreted in the subsequent sections.

4. Data Collection and Preparation:

This project data set was built based on various sources of Xoxoday related activity on twitter. The main elements consisted of a mentions-and-replies dataset with interactions among the users, a retweets dataset with amplification patterns, and node metadata with accounts of the participants of the interactions. All the datasets were cleaned and standardized in R using tidyr and dplyr packages and the janitor packages. The names of the columns were normalized, redundant fields were deleted and the missing values were addressed so that all the tables remained consistent. Directed interactions were represented by the use of the source and target nodes to form edges and node tables were built to distinguish unique users who were involved in the network.

Data preparation was performed by transforming raw text fields into the corresponding format, creating types of interaction, weighting edges between retweets, and eliminating unfinished or incorrectly constructed ones. This was done to make sure that every edge in the graph represented a valid interaction between two users and that all the nodes had been represented correctly. The resulting datasets were cleaned and then directed graphs were constructed that could be used to compute centrality, detect communities and be visualized.

Such procedural preparation afforded the correct and credible utilization of techniques of social network analysis.

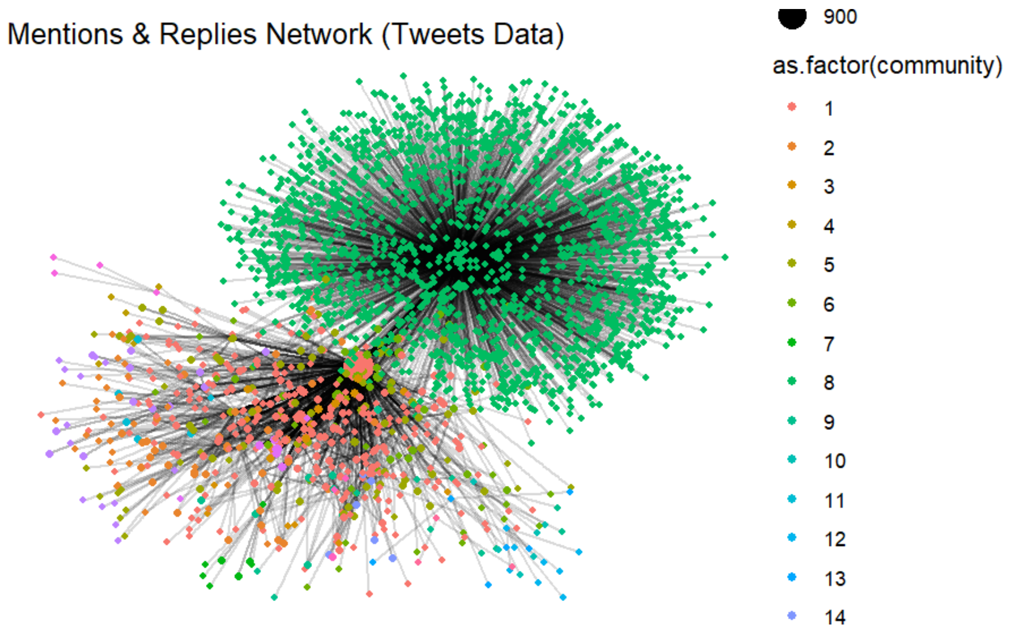
5. Methodology:

The following methodological approach utilized during this project is descriptive social network analysis which will be used to analyze the patterns of engagement in the Twitter activity of Xoxoday. Rather than describing interactions in unstructured ways, social network analysis offers a framework in which interactions are represented as a graph of nodes (users) and edges (tweets, mentions, replies, or retweets). The centrality measures were computed on four centrality measures: indegree, outdegree, betweenness, and PageRank. Indegree is one that identifies the most attention users whereas outdegree identifies the most initiating users. Betweenness centrality is the measure of the centrality of a user who can connect otherwise unconnected sections of the network and PageRank is the measure of the overall influence of user based on both direct and indirect connections. All of these measures can be seen as a display of the hierarchical order and power distribution in the network.

The Louvain modularity algorithm was used to do community detection by grouping together users who interact more with other members of their group than with the rest of users not subject to the group. This assists in exposing groups of talk and natural sub-communities. Temporal analysis was performed based on the frequency of tweets across time to find their pattern of activity and spike related to a campaign. The weights in retweets were summed to examine the strength of diffusion and which content was expanded the most. Reproducible R code was used to carry out all analyses, which could be evaluated in a transparent manner to determine engagement behavior.

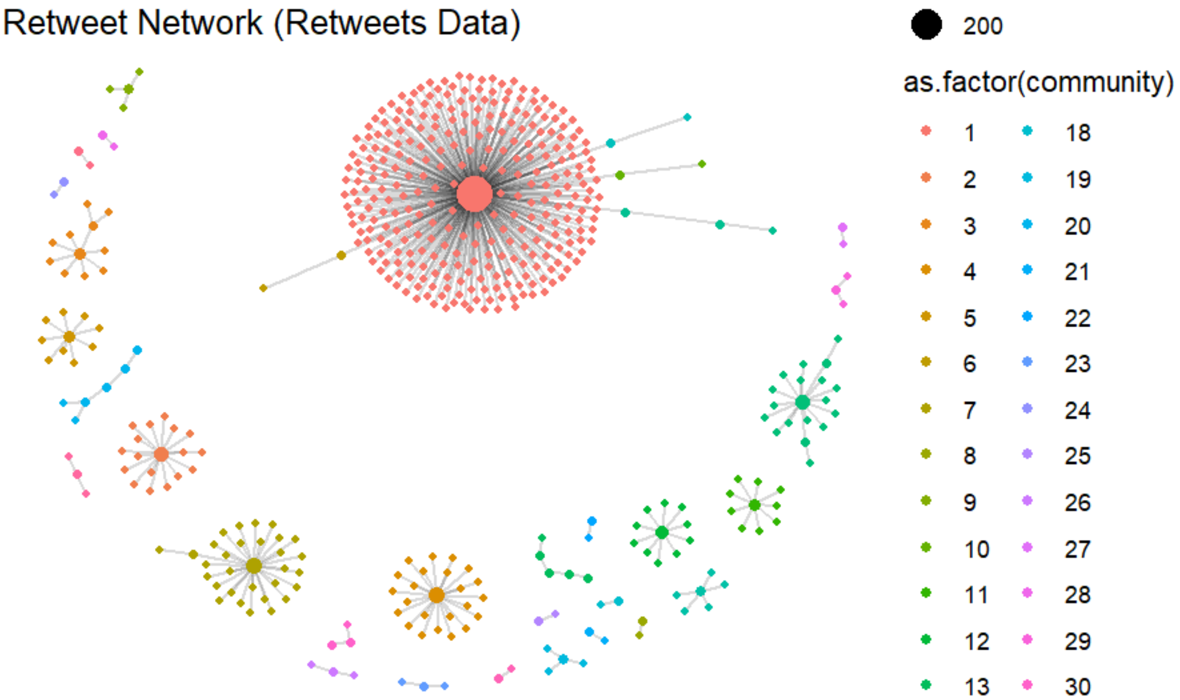
6. Network Visualizations and Interpretation:

6.1 Mentions & Replies Network Structure (Figure-1)



The Mentions & Replies network graph is a force directed layout, which depicts the interactions between users in terms of mentions and replies. The thick clusters and intersections indicate the way in which the discussions converge around several nodes. The most saturated and dark cluster is the one whose center is the account thexoxoday, and which involves the number of inbound edges forming a dense center of activity. The rest of the nodes in the surrounding extend out of this central node, uncovering a pattern of communication which heavily favors the brand account. The network is typical of a hub-and-spoke design, in which most users are in direct contact with the center, and not with one another. This implies a lack of peer-to-peer interaction and means that though Xoxoday is very prominent, the community does not portray a wide range of conversations.

6.2 Retweet Network Structure (Figure-2)



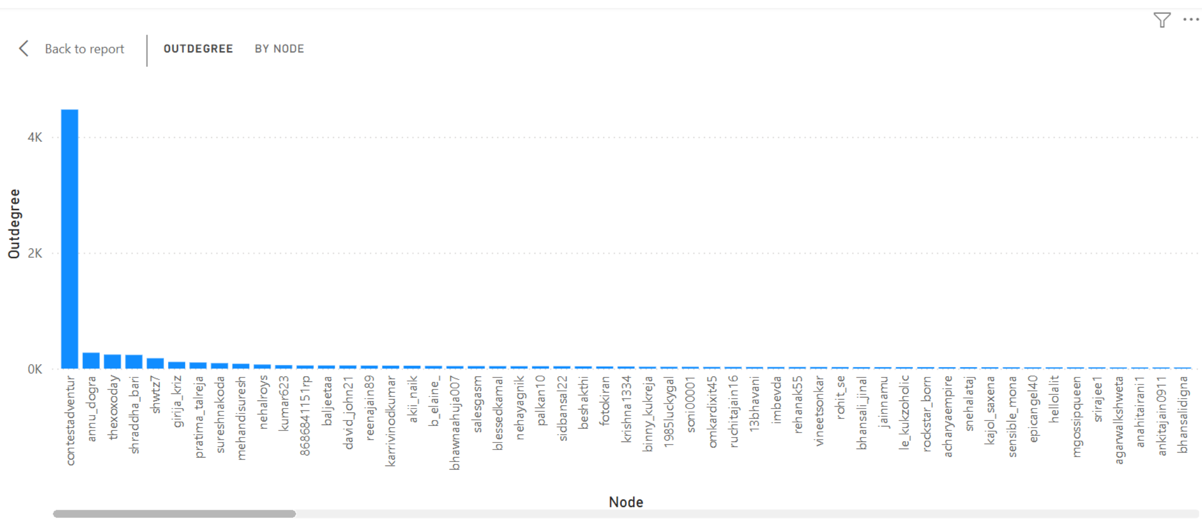
The Retweet Network graph also uses a force-directed layout but ends up giving it very different pattern compared to the mentions network. The retweet network is composed of numerous small, disjointed star-shaped subgraphs of individual original tweets, as opposed to two large clusters. The biggest starburst at the center is one tweet that had an unusual retweet rate and there are several retweets that then spread out in one layer. The other clusters are quite smaller in size and are mostly detached. The colored communities are groups that come up naturally as a result of Louvain community detection, but the groups are isolated events of diffusion as opposed to networked patterns of retweets. The architecture shows that the activity of retweeting is not widespread on the network, but instead event-focused.

6.3 Indegree Rankings (Mentions & Replies) (Figure-3)

Node	Indegree	Outdegree	IndegreeRank
thexoxoday	47	241	1
pratima_talreja	4	105	2
_ayushi_jain			3
deshapriya			3
_encounter			3
_gym_freak_		1	3
_indiandoc			3
_iwalk_alone			3
_joseph_vj			3
kkomal			3
_rashi__			3
_sir_deepak			3
_teehoo			3
_vgnsh			3
007preeti_			3
09sailesh			3
108pravi			3
Total			**3**

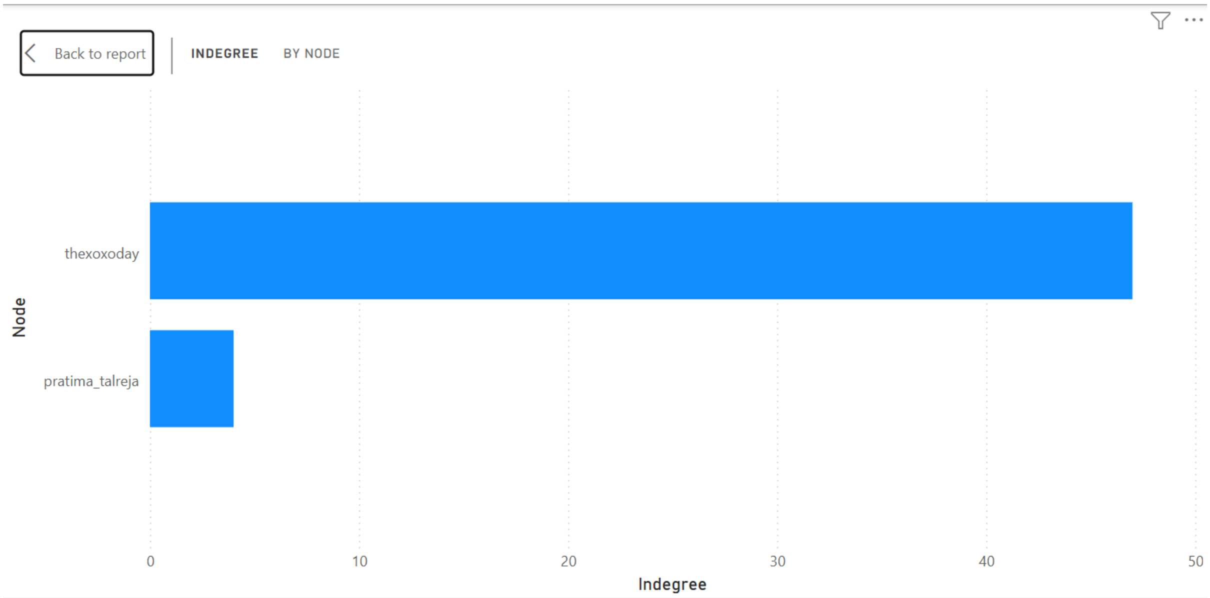
The indegree table contains the nodes with their indegree, outdegree and rank. The table validates the fact that thexoxoday has the highest number of mentions or responses with indegree of forty-seven. The second in order of account, called pratima_talreja has only four indegree, the rest having one or none. This numerical distribution highlights the centrality role of Xoxadays in the mentions network and shows the concentration of attention around the brand account. The network also exhibits minimal distributional conversational power, which implies that most of the communication is brand-based as opposed to community-based.

6.4 Outdegree Patterns (Mentions Sent) (Figure-4)



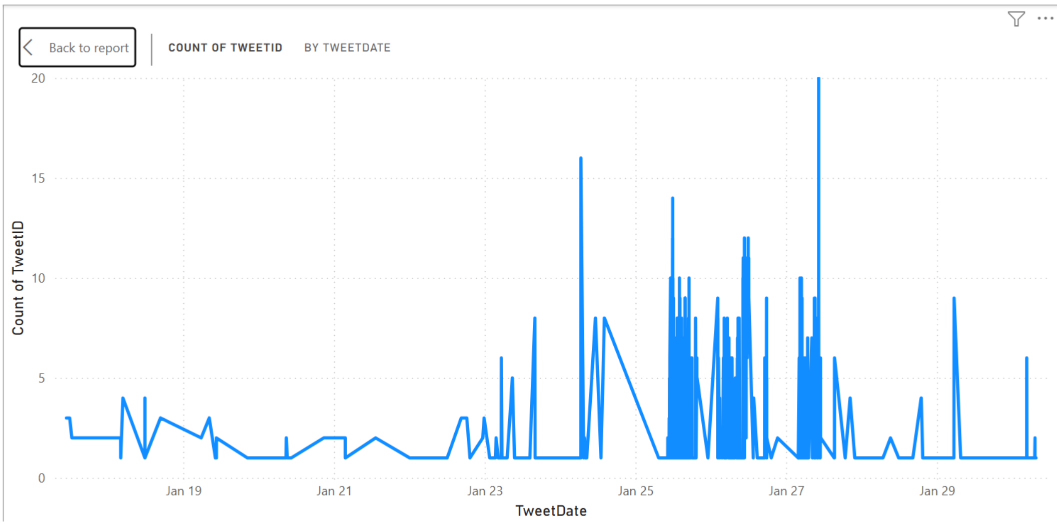
The bar chart demonstrates the quantity of outgoing interactions (mentions or replies) dispatched by every user. The salient characteristic is the excessive outdegree of the account, contestadventur (referring to which produces more than four thousand outgoing mentions) which is far beyond the outdegree of any other user in the dataset. This activity is aligned with fully automated tagging or contest promotional accounts. By comparison thexoxoday sends two hundred and forty-one outgoing interactions, which is also a significant number but much lower. A highly active contest account is an artificial boost to the apparent level of engagement of the network, in that it does not represent active conversation or community interest. This observation indicates that not every high-activity node can be of any value to genuine engagement.

6.5 Comparison of Top Indegree Nodes (Figure-5)



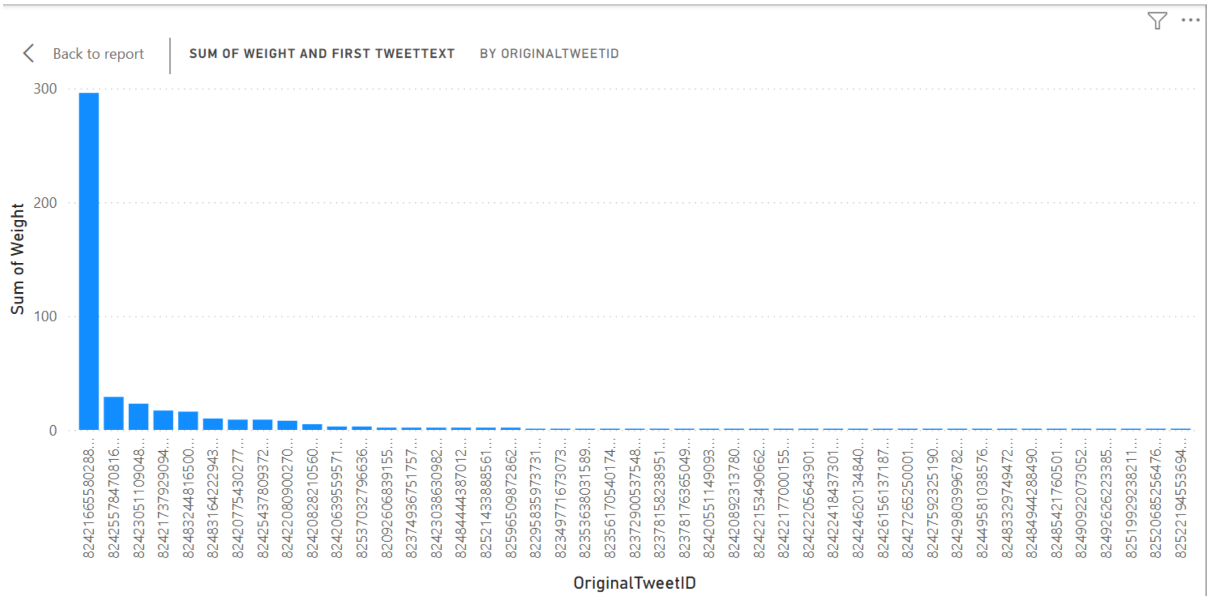
This is a visualization comparing the two with highest indegree values namely, thexoxoday and pratima_talreja. Thexoxoday has a domineering indegree of forty-seven whereas pratimatalreja is way below with four. The gap signifies the severe concentration of attention in the network. This visualization validates previous findings that the network has no influential or even strong secondary influencers or advocates whose interactions would diversify or enlarge the conversation. Rather, all the focus is absorbed on one testimony to the detriment of the capacity of the network to expand horizontally.

6.6 Tweet Frequency Over Time (Figure-6)



The time series chart shows the changes in the frequency of tweets throughout the time frame of analysis. In the earlier days, there is a low activity of one to three tweets daily. This becomes particularly active, though, in and about January twenty-sixth, but reaches its most active point on the twenty-seventh. This sharp rise would indicate an intensive campaign or special occasion which led to a more active use on the part of the users. The swift decrease in the following shows that the high activity was not persistent and a part of a long-lasting engagement tendency. The time series thus depicts a campaign based trend and not a steady organic interaction.

6.7 Retweet Weight Distribution (Figure-7)



The weight distribution of the retweets is the number of retweets each original tweet had. One tweet takes preeminence in the visualization, causing almost three hundred retweets. The number of retweets decreases drastically after that, and most of the tweets obtain low interest. The steep asymmetry shows that amplification of retweets relies nearly wholly on infrequent and extraordinarily successful posts. This trend indicates that even though Xoxoday can create viral content the most, most of the tweets fail to receive the same amount of traction and the amount of retweets is extremely uneven.

7. Integrated Interpretation:

All visualizations put together indicate that the Twitter network of Xoxoday is structurally centralized with thexoxiday acting as the central one. Competition-led action manipulates apparent action, especially by accounts whose outdegree patterns are extremely high and automated-like. The trend of retweeting is discontinuous and prone to individual viral moments and not through diffusion. Interactions over time are in tandem with the campaigns as opposed to a conversation. Collectively, the trends show that the presence of Xoxadays in social media is based on its core brand identity with minimal distributed community activity and regular organic content consumption.

8. Conclusion:

This project was aimed at learning about the way users use Xoxoday on Twitter and whether the network structure that is created supports organic and sustainable interaction. As the results show, Xoxoday is in a very central location in the network, as it receives most mentions and replies yet creates low levels of peer-to-peer interactions among the users. Tweeting is disaggregated and relies on a few high-performing tweets. The existence of the contest-based accounts also makes it harder to understand the engagement metrics because it introduces the artificially pumped-up activity, which does not indicate the interest of the audience in the issue. All these tendencies point towards the opportunities and threats of the social media strategy of Xoxoday.

The practice of carrying out this investigation made me realize how important it is to be careful with the data cleaning process, considerate of using network metrics, and careful in interpretations of anomalously active accounts. Provided that this research were extended, the further research could be the integration of other platforms, including LinkedIn or Instagram, or the comparison of engagement patterns of various types of campaigns. The other potential avenue would be to use predictive modeling to identify which content features have the strongest predictive power in terms of virality or engagement. On the whole, this review gives Xoxiday practical insights on how to enhance the development of its online communities and enhance the quality of interaction in all its social channels.

9. Reference:

- Borgatti, S. P., Mehra, A., Brass, D. J., & Labianca, G. (2009). Network analysis in the social sciences. *Science, 323*(5916), 892–895. https://doi.org/10.1126/science.1165821
- Berger, J., & Milkman, K. L. (2012). What makes online content viral? *Journal of Marketing Research, 49*(2), 192–205. https://doi.org/10.1509/jmr.10.0353
- Kwak, H., Lee, C., Park, H., & Moon, S. (2010). What is Twitter, a social network or a news media? In *Proceedings of the 19th International Conference on World Wide Web* (pp. 591–600). ACM. https://doi.org/10.1145/1772690.1772751
- Liu, Y., & Dai, R. (2020). Understanding promotional tagging behaviors in social media contests. *Social Media + Society, 6*(4), 1–12. https://doi.org/10.1177/2056305120970794
- Wasserman, S., & Faust, K. (1994). *Social network analysis: Methods and applications*. Cambridge University Press.